

# Code-Mixing and Code-Switching on Social Media

## Text: A Brief Survey

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**Abstract**—In the diverse linguistic landscape of India, individuals employ a variety of languages (codes) in their online communications, much like non-English speakers worldwide. Code-mixing is predominating in today's communication-driven culture and no natural language has escaped these trends. This paper explores the expanding subfield of Natural Language Processing (NLP) known as code mixing (CM) and code switching (CS). It examines how individuals exhibit their linguistic versatility by seamlessly blending English terms with regional languages (e.g., English-Hindi, English-Punjabi, English-Telugu) through techniques such as transliteration, Romanization, and language fusion on popular social media platforms such as Facebook, Google Reviews, and Twitter. The pervasive use of social media applications has resulted in a substantial influx of unstructured text, much of which is characterized by code-mixing and multi-script writing, where words are expressed in scripts distinct from their native tongue.

**Keywords**—Natural Language Processing, Machine Translation, Code-Mixing, Code-Switching.

### I. INTRODUCTION

Over the past decade, social media has witnessed an unprecedented surge in popularity, reshaping the dynamics of human interaction across various platforms. It is projected that by 2023, approximately 4.9 billion individuals worldwide will actively engage with social media platforms [1]. This transformative shift in the way people connect has been further fuelled by the advent of the internet and related technological advancements. As a result, fields such as anthropology, languages, and literature have witnessed the emergence of social computing studies, spanning a broad spectrum of cultural heritage, virtual environments, and e-culture. In a country like India that owes a rich tapestry of cultures and diverse populace, each region is characterized by its unique cultural identity. Platforms like Twitter, Facebook, and YouTube have become ubiquitous tools for individuals to share their thoughts and ideas, transcending geographical boundaries and reaching rural areas as well. A significant portion of India's rural population predominantly communicates in their mother tongues, given the limited

language, which is widely used for textual communication on social media. This linguistic preference allows them to express their opinions and connect with others, albeit creating a linguistic barrier for those from different cultural or linguistic backgrounds. The proliferation of social media content has brought forth both new opportunities and challenges in understanding culture and accessing information.

### II. CODE-SWITCHING AND CODE-MIXING

Code-Switching happens when a speaker switches between two or more languages according to their respective cultures or linguistic variations in the course of a single conversation or circumstance [2]. Census data indicates that slightly over 25% of the population is bilingual, while only 7% are trilingual. In multilingual interactions, individuals often combine elements from multiple languages. This phenomenon is most commonly observed when bilingual speakers seamlessly blend their languages, alternating between them within a single sentence or phrase. A few examples of various forms of code switching are shown in Table I.

TABLE I  
EXAMPLES OF CODE SWITCHING

| Code | Sentence                                      | Code Switching Type      |
|------|-----------------------------------------------|--------------------------|
| CS1  | “Yeh Dil Maange More!”                        | tra-Sentential Switching |
| CS2  | ome on girls, Waqthai shine karne ka!”        | tra-Sentential Switching |
| CS3  | “Piyo glass full doodh, dhooth hai wonderful” | Tag Switching            |
| CS4  | “All backbenchers are chichorass”             | Intra-Word Switching     |
| CS5  | “Hayeee, How are you”                         | Intra-Word Switching     |

Code mixing is a closely related phenomenon to code-switching, involving the integration of words or phrases from one language into another within a single sentence, a process referred to as Code Mixing [3]. This natural occurrence often stems from the use of multiple languages and occurs seamlessly without a shift in the conversation's topic, manifesting across various linguistic levels.

When a word or phrase from another language is incorporated into a sentence or sentence fragment typically expressed in the local language, this linguistic phenomenon is termed 'intra-sentential code mixing.' Conversely, when an individual transitions from one language to another within the same discourse or utterance, this is referred to as 'inter-sentential code mixing.' Moreover, the practice of code mixing within a single word is known as 'intra-word code mixing.' A few examples of various forms of code mixing are shown in Table II.

TABLE II  
EXAMPLES OF CODE MIXING

| Code | Sentence                                           | Code Mixing                   |
|------|----------------------------------------------------|-------------------------------|
| CM1  | "NaNa seriously, 'Mein Delhi mein nahi tha'"       | Intra-sentential code mixing  |
| CM2  | "Amit sir ki jacket ka color nice hai "            | Intra-sentential code mixing  |
| CM3  | "r u interested, 'chalo is k bhi naam likho'"      | Inter- sentential code mixing |
| CM4  | "Aaj ka din rahe ga b'cos get placement in Google" | Inter- sentential code mixing |
| CM5  | " You know , 'Bindi is my favourite sabji'"        | Intra-word code mixing        |

### III. RESEARCH ISSUES AND CHALLENGES

The frequent language switching observed among multilingual and bilingual speakers poses a significant challenge for the automated interpretation of social media text. This challenge arises from various factors, including emerging linguistic trends, modernization, or writing styles, that may result in incorrect spellings or interpretations. Users further contribute to this challenge through intentional lengthening and shortening of words as given ahead.

#### A. Prolonged words used intentionally

Users often elongate the haracters included within words to show their feelings of enthusiasm. Example: - helllllloooooo, thankuuuuu etc.

#### B. Shortening of words

In mixed data, creative spelling and word shortening are also common, as exemplified by terms like 'gudluk,' 'gr8,' 'tc,' 'pix,' 'admin,' and '2day,' which further complicates the interpretation of such content.

#### C. Phonetic misspellings

In modern English text, phonetic spellings based on the local language are common. These phonetic misspellings are often understood by those familiar with the language. For instance, the substitution of '4' for 'for' results in 'formal' being written as '4mal,' and similarly many such instances occur.

#### D. Multiword tokens and abbreviations

Users frequently employ acronyms and multiple-word abbreviations, known as multiword tokens, which require regular updates due to their expanding volume. For example, 'JK' stands for 'Just kidding,' and 'BF' is an abbreviation for 'Best friend.'

#### E. Ambiguous words

Some words used in social media text have multiple meanings in different languages, stemming from phonetic typing. These ambiguous words may have contrasting interpretations in various languages. For instance, 'bus' means 'stop' in Hindi and 'large vehicle' in English, and 'din' signifies 'day/time' in Hindi and 'noise' in English.

#### F. Named entities

Numerous individuals have names that resemble movie titles, places, or common items, leading to potential ambiguity in their interpretation. For instance, in the sentence 'Lava ke mobile light weight hote hai,' 'lava' could refer to both a brand or hot liquid rock.

#### G. Hinglish words

A wide array of words is formed by combining English words with Hindi suffixes. These Hinglish words are created by using English as a base and adding Hindi suffixes. Examples include 'madamji' and 'bottloan,' which denotes 'many bottles.'

#### H. Reduplication

Reduplicated words are utilized in both Hindi and English. In Hindi, some words are repeated or modified, contributing to potential interpretive challenges. For instance, words like 'aisa-waisa,' 'Kit-kat,' 'super-duper,' and 'kal-val' are frequently encountered.

#### I. Same words across languages

Several English words, such as 'jungle,' 'verandah,' 'thug,' 'scooter,' 'T.V.,' 'radio,' 'department,' 'video,' 'cigarette,' 'diploma,' 'flashback,' 'furniture,' and others, are commonplace in Hindi, leading to potential cross-linguistic ambiguity.

### IV. RELATED WORK

Numerous researchers have explored the factors contributing to code mixing on social media. In this section, we delve into

the relevant research within the domains of code switching and code mixing. These complementary approaches collaborate to enhance the accuracy and utility of translation applications across diverse languages and text sources [32][33].

Related works in this direction and their prominent features, as collected from the explored literature over the last two decades, have been summarized in Table III and is as given below:

TABLE III.  
RELATED WORKS

| Sr. No. | Authors [Reference]         | Year | Publishers           | Salient Feature of the Work                                                                                                                                                                                                                                                                              |
|---------|-----------------------------|------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Sinha R.M. et al. [4]       | 2005 | ACL                  | A system for the translation of Hinglish automatically into Hindi and English forms. Developed methods for identifying them and translating them into pure Hindi and pure English forms [4].                                                                                                             |
| 2       | Aw A. et al. [5]            | 2006 | ACL                  | A phrase based system for normalizing SMS mobile phone texts which included (Short forms, creative or stylistic spelling, punctuation omission) [5].                                                                                                                                                     |
| 3       | Choudhury M. et al.[6]      | 2007 | Springer             | Develop a word-model based on a hidden markov model for texting language after researching the types and nature of compressions used in SMS texts [6].                                                                                                                                                   |
| 4       | Solorio T. et al. [7]       | 2008 | ACL                  | Developed a POS tagger for the identification of word from English-Spanish code-switched text [7].                                                                                                                                                                                                       |
| 5       | Cook P. et al. [8]          | 2009 | ACL                  | Proposed unsupervised noisy channel model for normalization of data [8].                                                                                                                                                                                                                                 |
| 6       | Pavan K. et al. [9]         | 2010 | Macmillan Publishers | A recommendation for the Romanized text language identification system (RoLI) that takes these concerns into account. RoLI employs both a sound-based similarity of words and an n-gram method [9].                                                                                                      |
| 7       | Clark E. et al. [10]        | 2011 | ELSEVIER             | They provide that non-native English speakers, elderly Internet users, and others who are not part of the "in-group" frequently have trouble understanding such texts which explores the challenges of automatically standardizing social media English [10].                                            |
| 8       | Sharma S. et al. [11]       | 2012 | Coimbatore IIT       | Proposed phrase-based statistical MT technique. For the English-Hindi language pair with Indian names, machine transliteration is using UTF and wx-notation for different character encoding for the target language [11].                                                                               |
| 9       | Sequiera R.D. et al. [12]   | 2014 | ACM                  | The proposed model will take as input a sequence of Romanized words and output the corresponding languages and their native script transliteration [12].                                                                                                                                                 |
| 10      | Sharma S. et al. [13]       | 2015 | IEEE                 | Determined the language used in code-mix texts, which consists of slang, abbreviations, word play, and phonetic typing [13]. They transliterated these Romanized English terms, which are spelled differently, and used a lexical technique to determine whether the statement was positive or negative. |
| 11      | Dutta S. et al. [14]        | 2015 | IEEE                 | Discussed about English spelling mistakes were found in social media posts that included both Romanized transliterations [14].                                                                                                                                                                           |
| 12      | Ranjan P. et al.[15]        | 2016 | IEEE                 | Presented the degree of code mix in the corpora that bootstrap the language identification phase and compute the Code Mixed Index [15]. They build a language model using pen tree bank data and code mixed data using the RNNLM                                                                         |
| 13      | Sharma V.K. et al.[16]      | 2016 | ELSEVIER             | Explores the use of the Wikipedia API for cross-language information retrieval between Hindi and English that can retrieve information in both Hindi and English languages using a single search query [16].                                                                                             |
| 14      | Phadte A. et al. [17]       | 2017 | ACM                  | Tested language detection at the word level using the Konkani-English tagged dataset [17].                                                                                                                                                                                                               |
| 15      | Lakshmi B.S. et al.[18]     | 2017 | IEEE                 | Emphasized on the word-level LID for code mixed data issue. Sentences from social media posts that are combined with English and Kannada and applied different supervised classification techniques for LID [18].                                                                                        |
| 16      | Jamatia A. et al. [19]      | 2018 | Springer             | POS tagging using coarse-grained and a fine-grained tag set [19], and compare the complexity with other code-mixed corpora.                                                                                                                                                                              |
| 17      | Dhar M. et al.[20]          | 2018 | ACL                  | Described the design of Hinglish code-mixed text and offered an augmentation pipeline method to complement traditional machine translation (MT) techniques that can aid in producing better translations [20].                                                                                           |
| 18      | Mave D. et al. [21]         | 2018 | ACL                  | Presented Code switching behavior, the data source, the morphology of the languages involved, and their relationships to one another all affect how difficult it is to identify the language in code switched data [21].                                                                                 |
| 19      | Sharma S. et al. [22]       | 2018 | Springer             | Cultural computing methods from a variety of angles and perspectives [22]. The poll emphasizes the need for cultural computing to be integrated into the social computing space already in place, or vice versa.                                                                                         |
| 20      | Sreelakshmika K. et al [23] | 2019 | ELSEVIER             | Presented a machine learning model for detecting hate speech in code-mixed social media text in Hindi and English [23].                                                                                                                                                                                  |

|    |                           |      |                      |                                                                                                                                                                                                                                                                                                              |
|----|---------------------------|------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21 | Shekhar S. et al. [24]    | 2019 | IOS Press Netherland | Determine Hindi-English code-mixed data used on social media platforms [24] and recommend a deep learning framework based on cBoW.                                                                                                                                                                           |
| 22 | Bhattu S.N. et al. [25]   | 2020 | ACM                  | Developed a method for text scraping and producing word embeddings. They constructed a hierarchical deep RNN with a linear-chain CRF layer to capture numerous contextual word properties [25].                                                                                                              |
| 23 | Vathsala M.K. et al. [26] | 2020 | Springer             | Presented code-switched data and transliterated to English by the use of LSTM and RNN network [26]. Tensor Flow is used to express LSTM appropriately during the course of the work.                                                                                                                         |
| 24 | Dowlagar S. et al. [27]   | 2021 | ACL                  | Proposed the language identification and POS tagging structure for the code-mixed text [27]. A Transformer with convolutional neural network architecture was employed. It uses code-mixed social media material to build a joint learning technique that combines POS tagging and LI models.                |
| 25 | Gautam D. et al. [28]     | 2021 | ACL                  | The Authors proposed mBART technique, a sequence-to-sequence model i.e multilingual, that has been previously instructs to transliterate romanized Hindi words into Devanagiri script [28].                                                                                                                  |
| 26 | Jadhav I. et al. [29]     | 2021 | IEEE                 | Designed a pipeline, which improves Hindi-English code-mixed text translation using a language identification layer was developed, and then, it was evaluated against the current Google neural machine translation system [29].                                                                             |
| 27 | Mundra S. et al. [30]     | 2022 | Springer             | Presented hate speech and offensive text classification in Hindi and English (Hi-En) [30]. They used various approaches like stack CNN and LSTM to extract local and sequential semantic features. Code-mixed hybrid embedding (CMHE) highlights one of the issues of variation in spelling and code-mixing. |
| 28 | Patil A. et al. [31]      | 2023 | IEEE                 | Focus on improving the capability of code-mixed NLP tasks [31]. Presents an assessment of different pre-trained unsupervised Transformer-based language models.                                                                                                                                              |

## V. CONCLUSION

The amalgamation of syntax and lexicon from two or more languages leads to a significant rise in the creation of novel phrases. Code mixing presents a formidable challenge for researchers working in this field. This available work emphasizes the utilization of Natural Language Processing (NLP) methods and Machine Learning (ML) algorithms for effectively processing code-mixed data. The research conducted thus far has addressed a spectrum of challenges, including machine translation, dialect identification, parsing, and automatic speech recognition (ASR). Furthermore, certain studies have delved into the complexities of bilingual datasets. These investigations collectively contribute to a deeper understanding of the intricate landscape of code mixing and its implications in the realm of language processing and analysis.

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